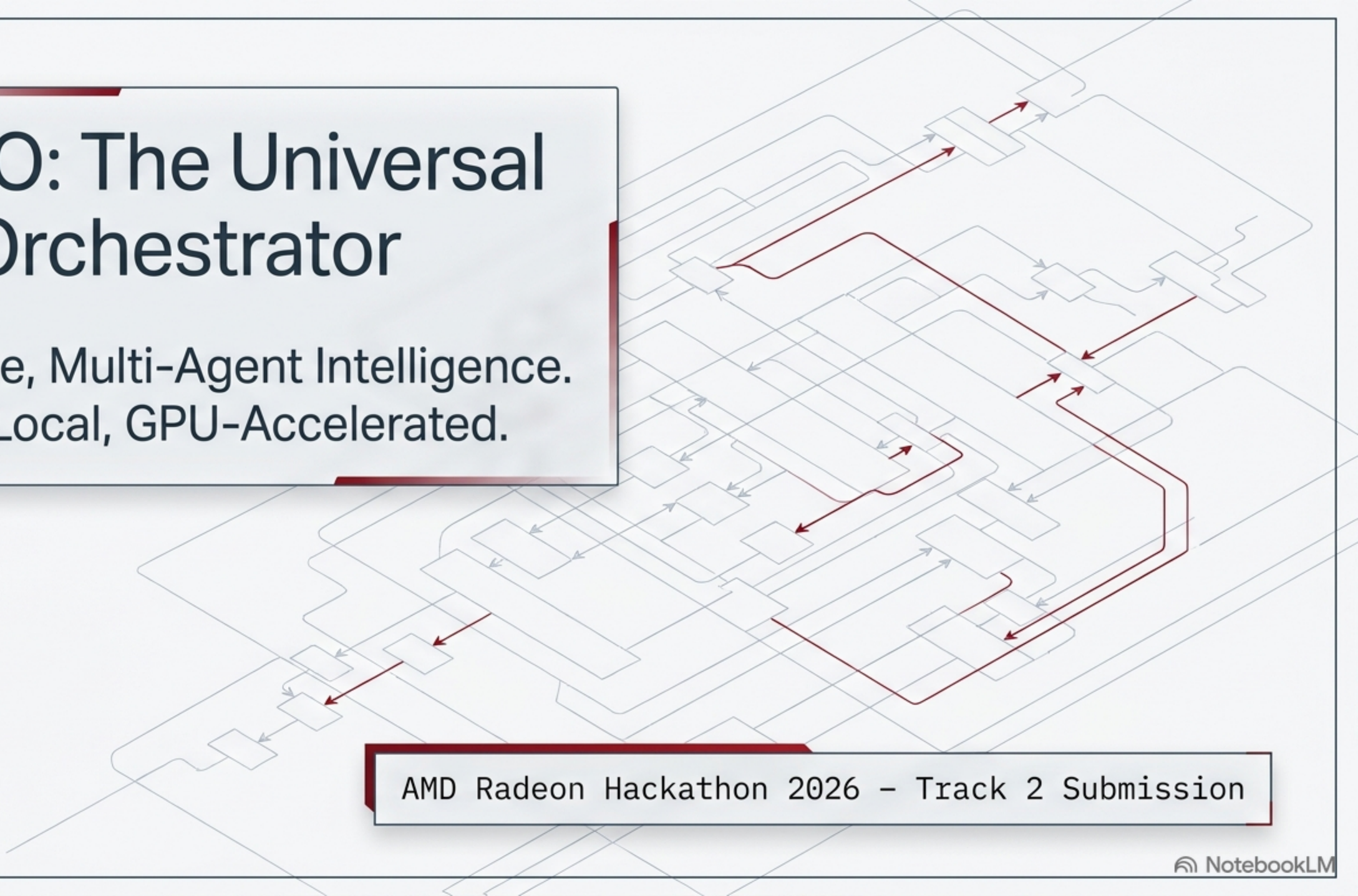
MKO: The Universal AI Orchestrator

Private, Multi-Agent Intelligence.
Fully Local, GPU-Accelerated.

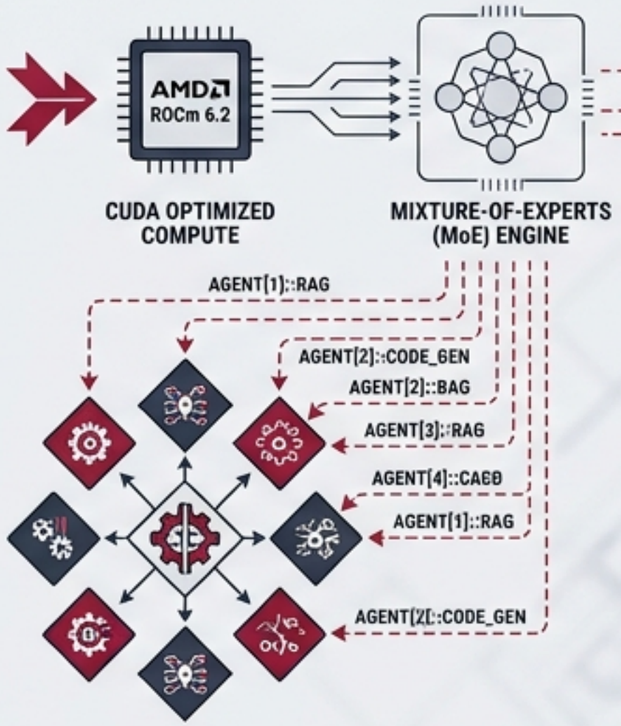


AMD Radeon Hackathon 2026 – Track 2 Submission

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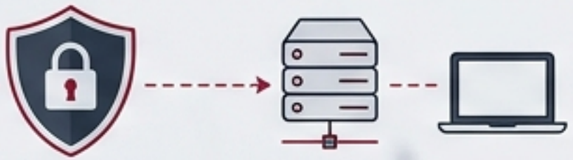
Three core philosophies drive MKO's architecture

Power (Hardware Accelerated)

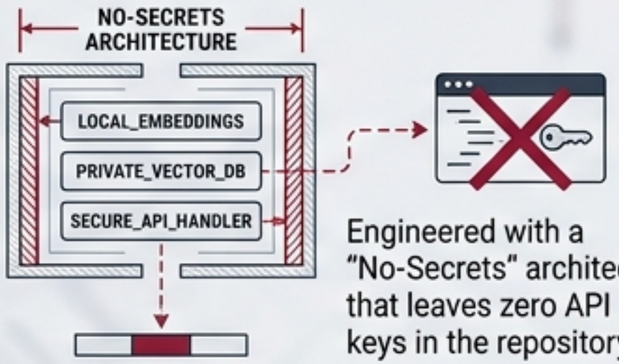


Powered by a dynamic Mixture-of-Experts (MoE) engine routing to 8 specialized agents.

Privacy (Zero-Leak Design)



100% local runnability.
On-device RAG document processing.



Engineered with a "No-Secrets" architecture that leaves zero API keys in the repository.

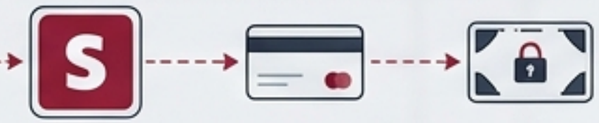
Polish (Enterprise UX)



Multi-provider streaming via Server-Sent Events (SSE).



Real-time SVG telemetry visualizations.



Built-in Stripe monetization system.

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A production-ready stack optimized for latency and local control

4. Client / UX

Vanilla JavaScript (ES Modules), CSS3, SVG.
Zero framework bloat for maximum render speed.

3. Backend Engine

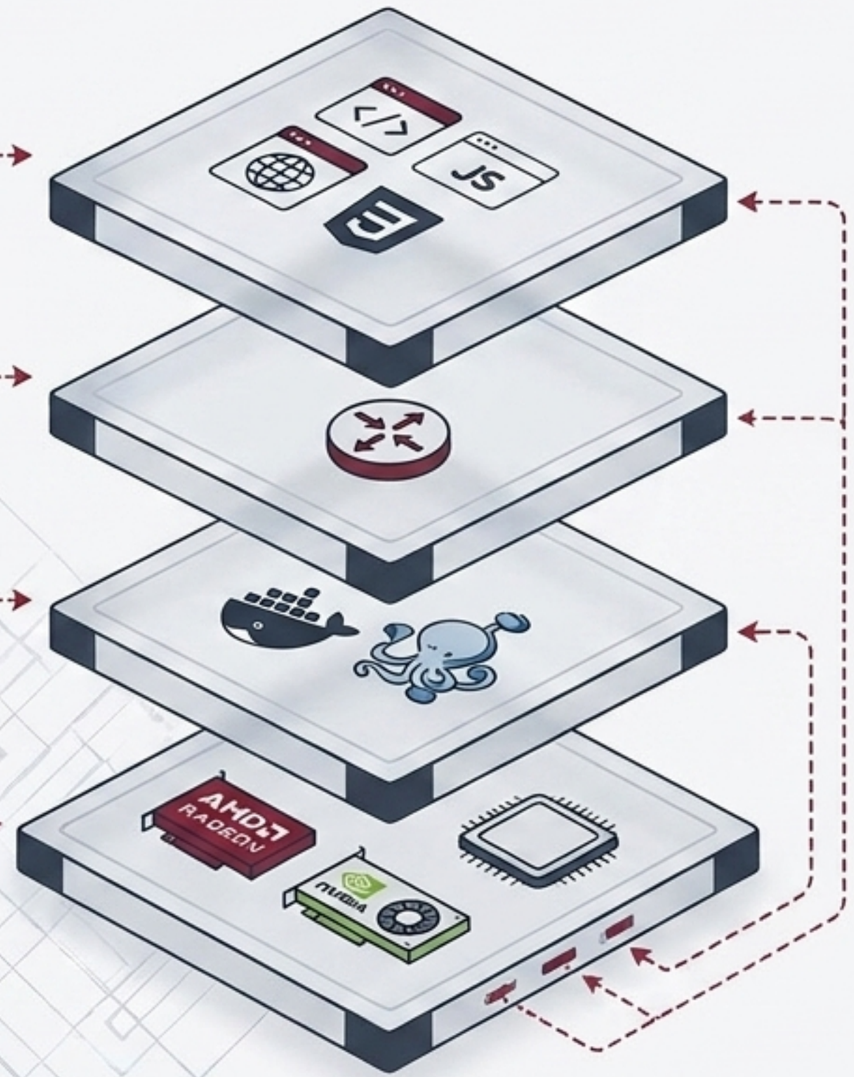
Python 3.10+, FastAPI, PyTorch, Uvicorn.
Powered by the unified UniversalLLM router.

2. Deployment

Docker & Docker Compose.
One-command infrastructure bring-up.

1. Hardware

AMD Radeon GPUs / NVIDIA GPUs / CPUs.
Native RDNA3 support via ROCm 6.2.



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Eight specialized cognitive agents handle distinct workflows

Cognitive Core

General
All-purpose assistant for conversation and Q&A.

Reasoner
Logical analysis and critical thinking across multiple perspectives.

Workflow & Task

Planner
Breaks tasks into actionable steps with time estimates.

Actor
Workflow automation generating strictly structured outputs.

Information & Data

Researcher
Deep-dive analysis yielding structured findings and sources.

RAG Agent
Contextual generation from local Qdrant-indexed documents.

System & Routing

Memory
Contextual recall and persistent cross-conversation state tracking.

MoE Agent
Parallel LLM dispatch with weighted response synthesis.

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The UniversalLLM router seamlessly bridges local and cloud inference



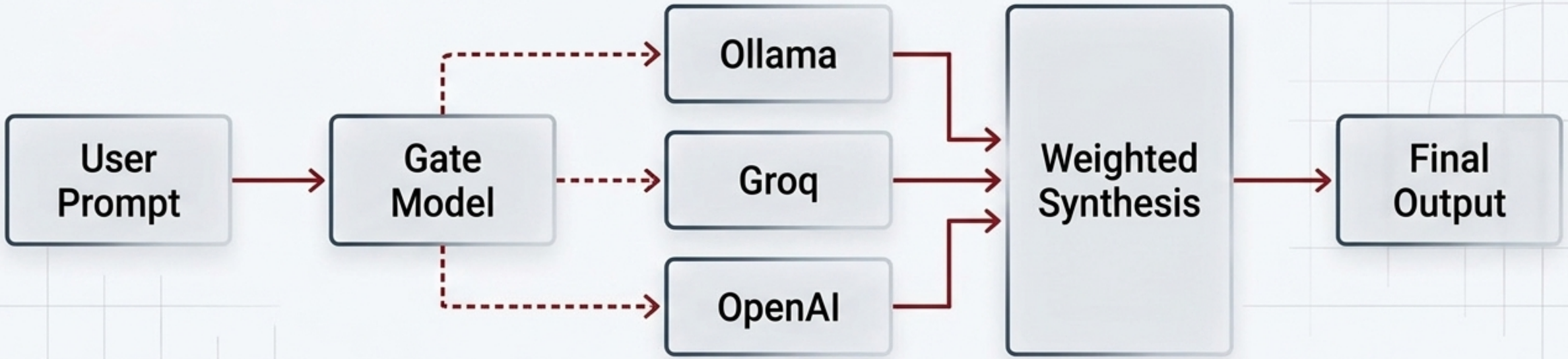
Provider	Type	Default Model	Free Tier Status
Ollama	Local	`llama3.2`	**Always Free**
Groq	Cloud	`llama-3.3-70b-versatile`	Free Tier Available
HuggingFace	Cloud	`zephyr-7b-beta`	Free Tier Available
OpenAI	Cloud	`gpt-4o-mini`	Paid
Anthropic	Cloud	`claude-3-5-haiku-20241022`	Paid

Graceful Degradation

All providers stream token-by-token over SSE. Missing API keys or an offline Ollama server yield friendly UI warnings mid-stream, never an application crash.

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The Mixture-of-Experts engine synthesizes parallel intelligence



Step 1: Configure & Weight

User selects any combination of expert providers and assigns influence weights via the UI.

Step 2: Parallel Dispatch

Prompt dispatched simultaneously to all selected experts utilizing Python's `asyncio.gather`.

Step 3: Execution & Telemetry

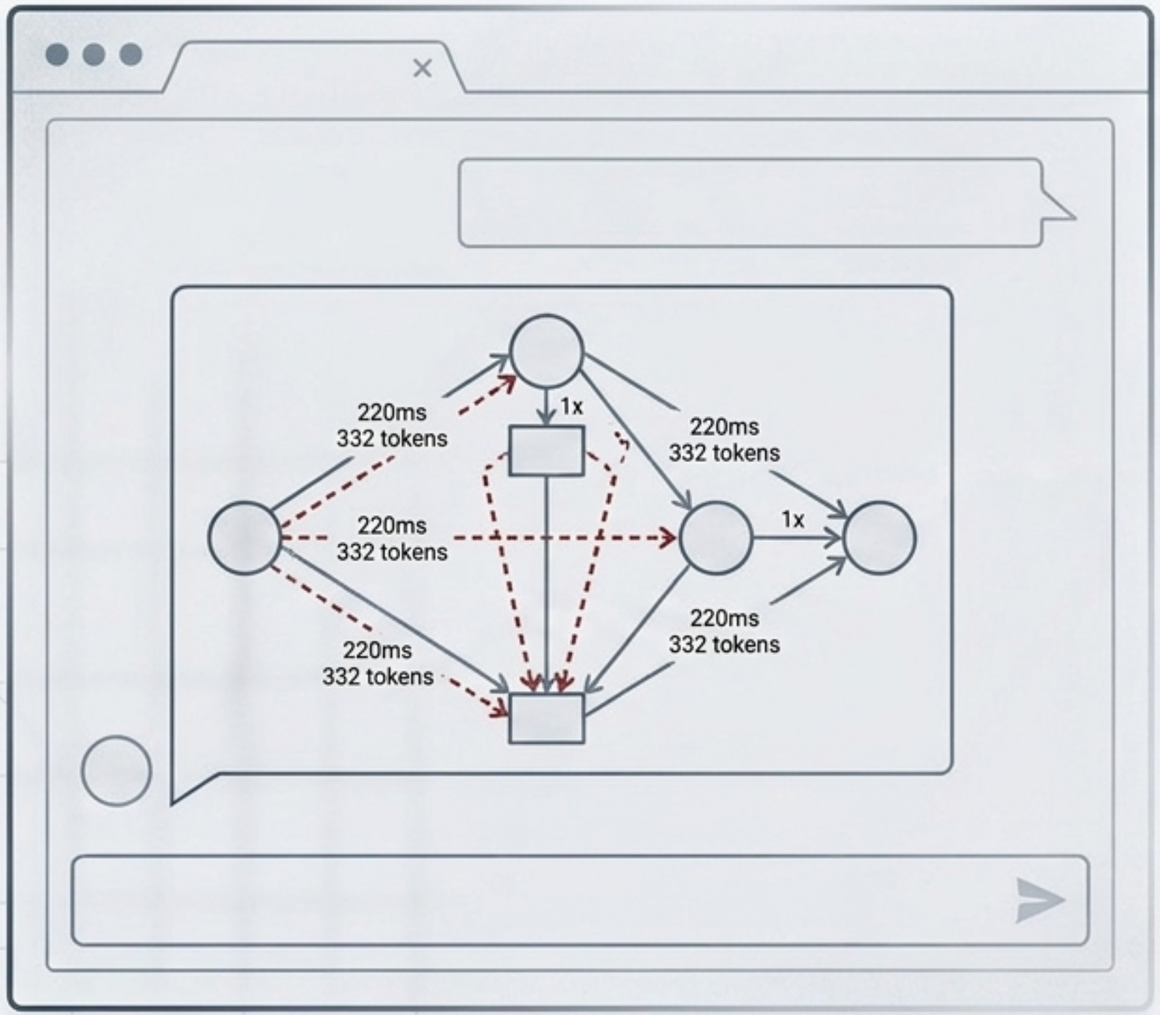
Each model streams its response alongside real-time latency and token metrics.

Step 4: Weighted Synthesis

Gate model intelligently merges disparate expert outputs based on configured weights to generate the final optimal response.

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Real-time observability rendered with zero framework dependencies



Pure Vanilla Execution

DAG layout logic (~290 lines) and rendering rely exclusively on pure JS and inline SVG elements. No React, no D3.js.

Live Edge Telemetry

Graph edges dynamically update with expert weight, latency, and token counts as SSE events arrive.

Deterministic Aesthetics

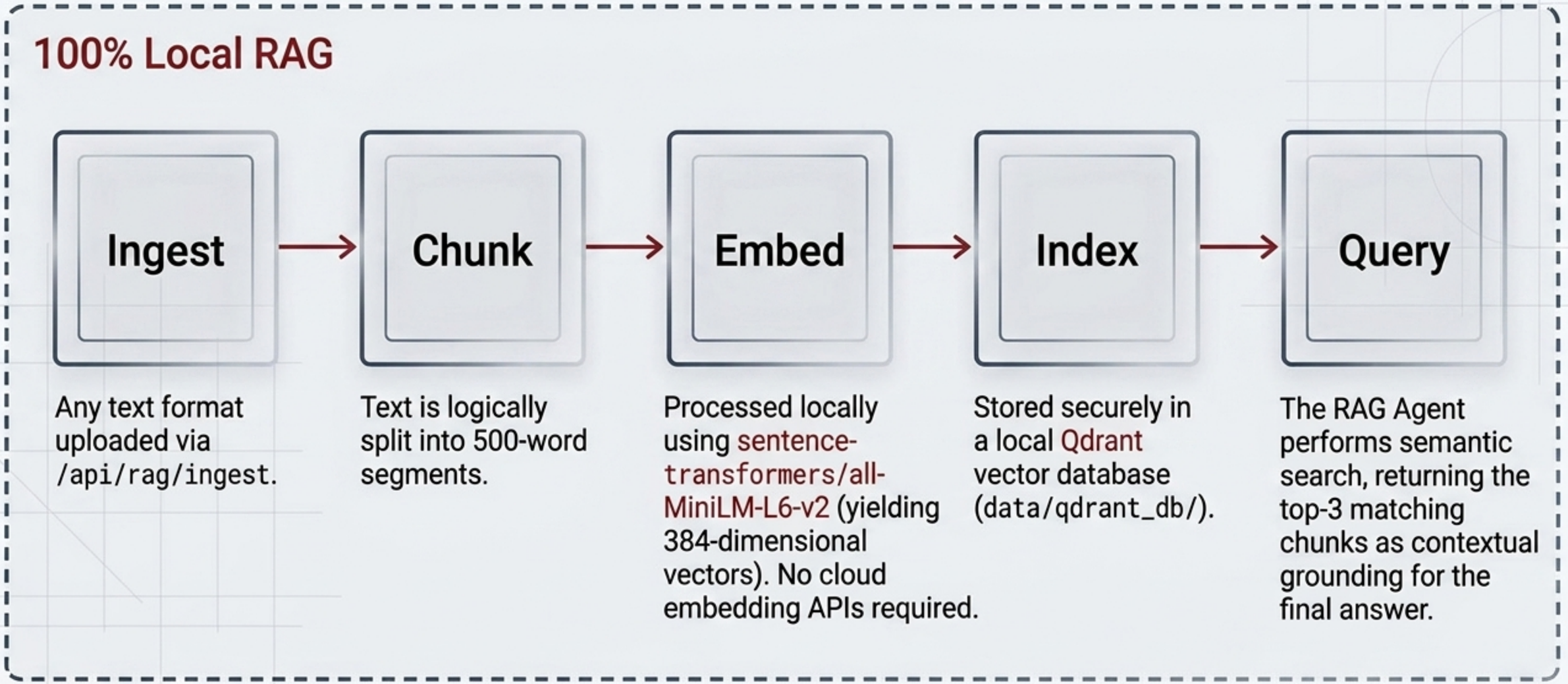
Graph colors are auto-generated consistently using a DJB2 hash of the provider's name.

Screen-Reader Accessible

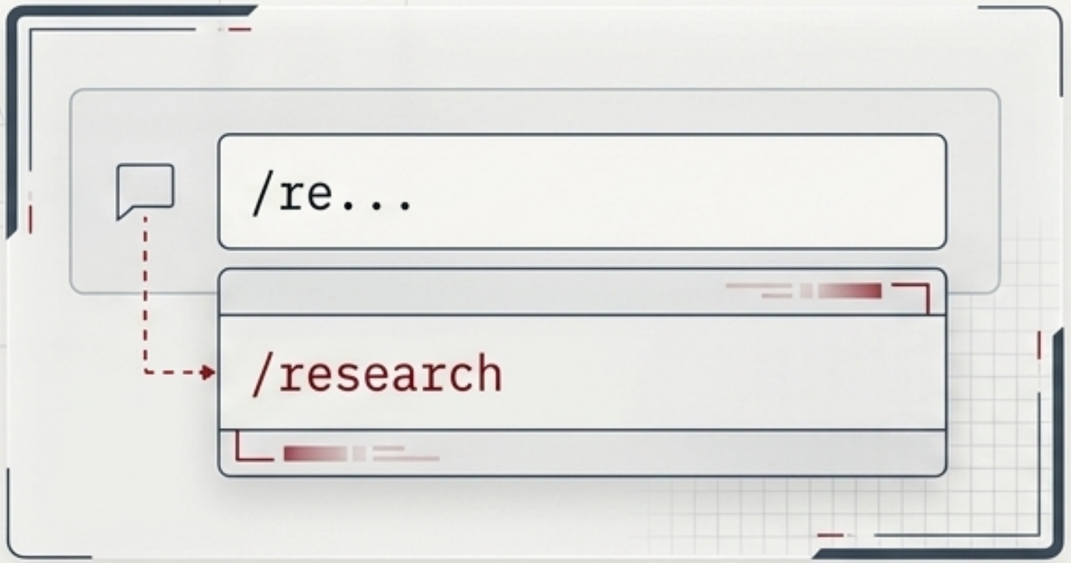
Semantic titles and descriptions ensure telemetry remains fully accessible.

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Private enterprise knowledge stays entirely on-device



Fluid user experience powered by rigorous test coverage

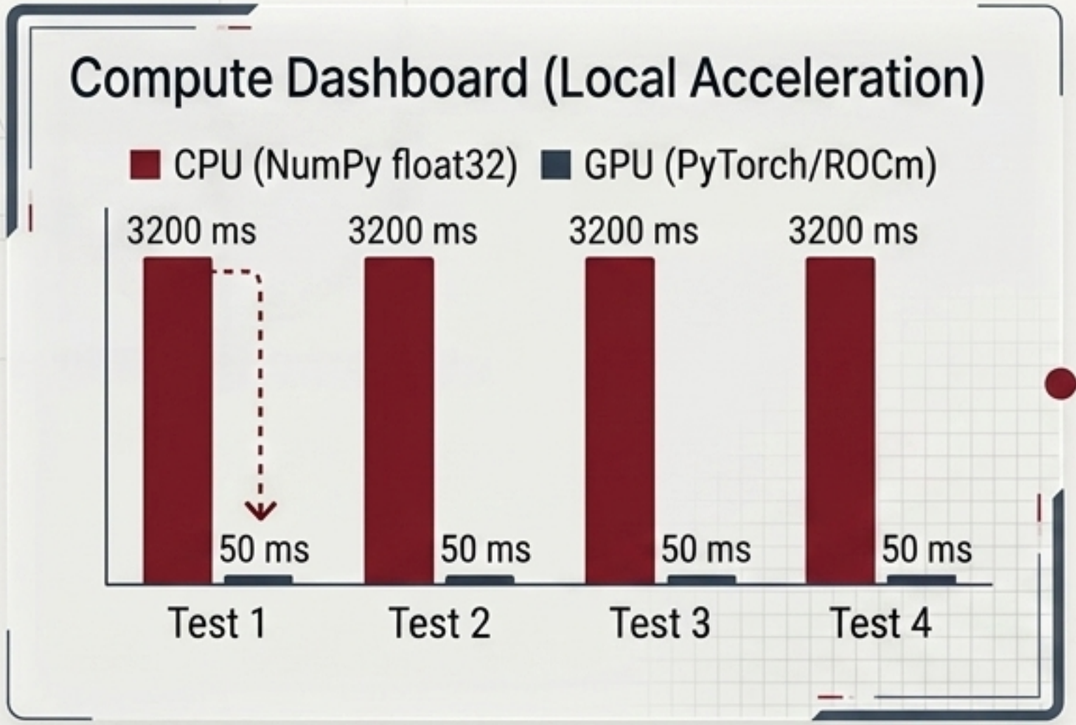


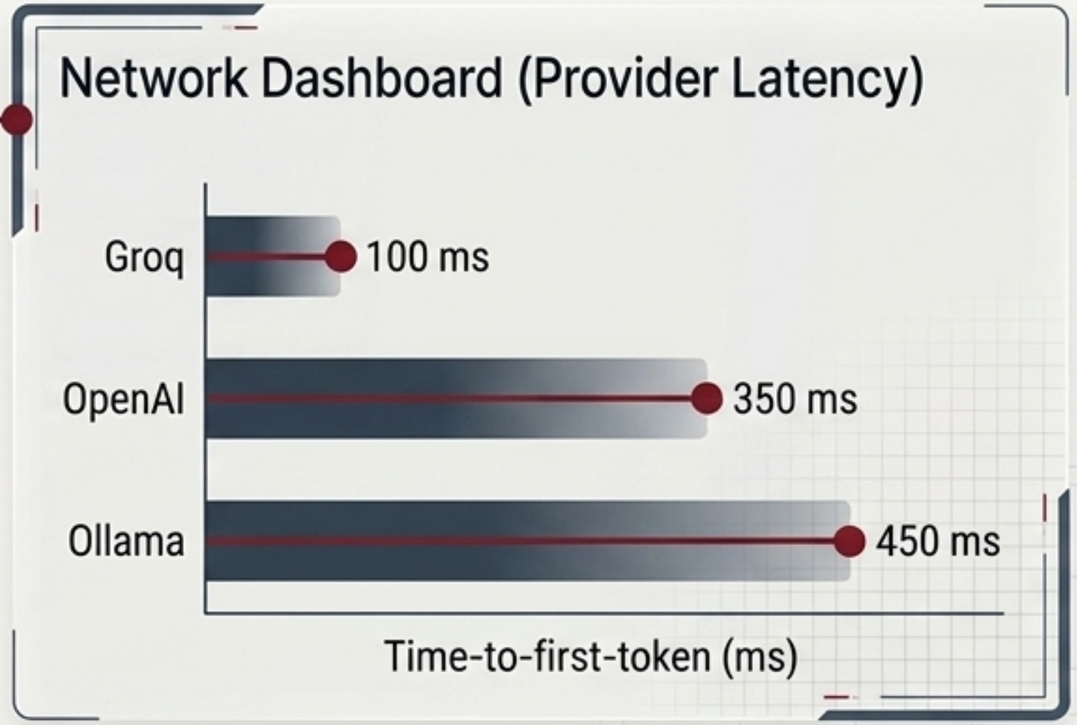
- Instant navigation: Type / to swap agents (`/plan`), switch models (`/model groq`), or view dynamically generated help (`/help`).
- Live keyboard-navigable autocomplete suggestions.

Vanilla JS Module	Built as a pure `Vanilla JS UMD` module.
Test Coverage	Backed by **47 distinct `Node.js` native test cases**.
Execution Rules	Enforces **case-insensitivity**, **trailing slash stripping**, and **intelligent error bubbling** for unknown commands.

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Built-in hardware benchmarking proves ROCm GPU acceleration





`POST /api/benchmark/run`. Compares CPU (`NumPy float32`) against GPU. Calculates matrix multiplication speedups on matrices up to `3000x3000`, deriving `real-time estimated LLM tokens/sec`.

`GET /api/benchmark/providers`. Tracks status, `time-to-first-token` (`ms`), and total time to identify network bottlenecks.

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Enterprise-ready administration and built-in monetization

Stripe Credit System

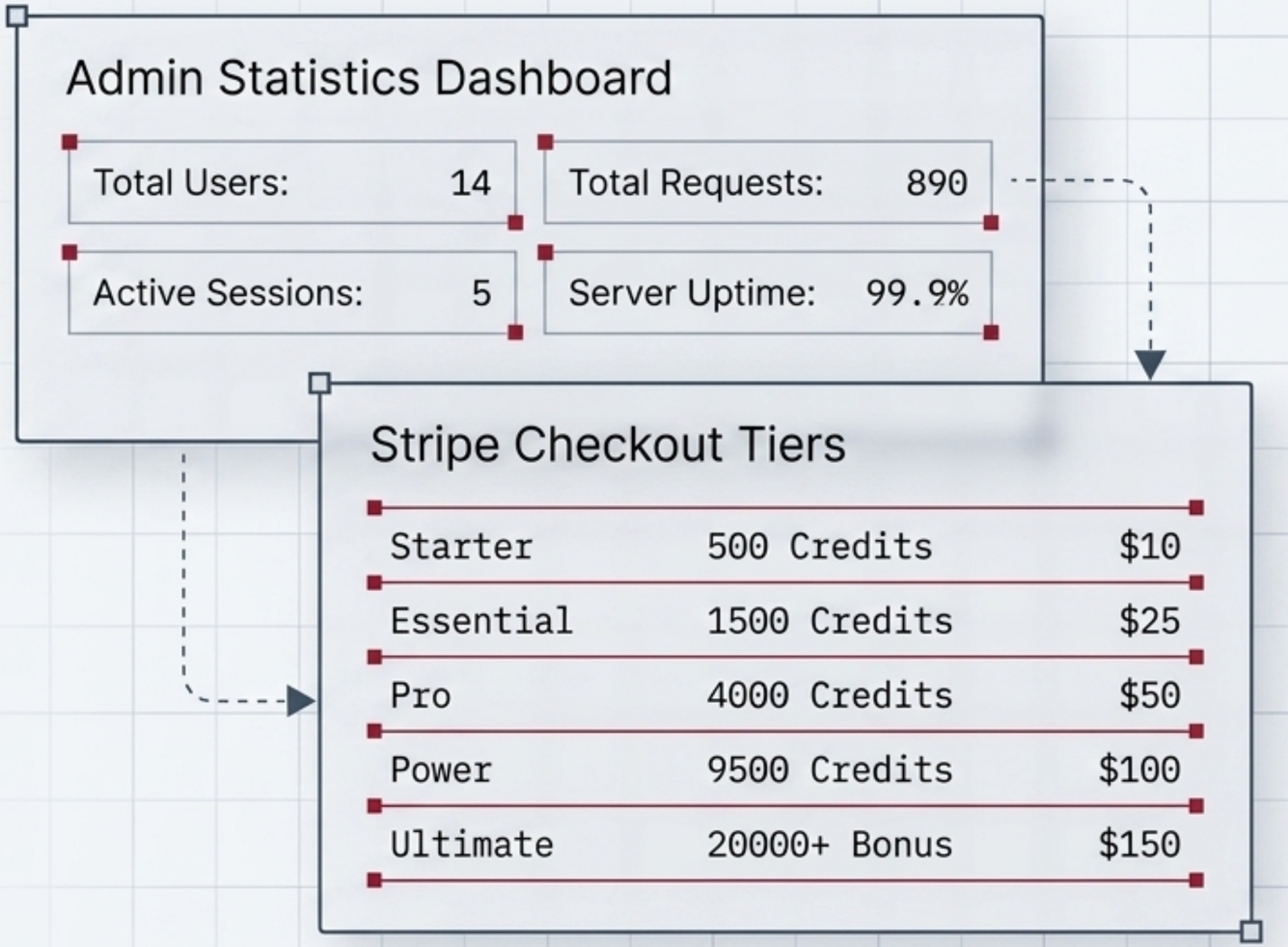
Fully integrated Stripe Checkout supporting 5 distinct tier packs (Starter: $10 for 500c up to Ultimate: $150 for 20k+ bonus).

Offline Demo Mode

Functions seamlessly without Stripe API keys using 6 built-in redemption codes (e.g., MKO-POWER-9500) to enable local evaluation.

Admin Dashboard

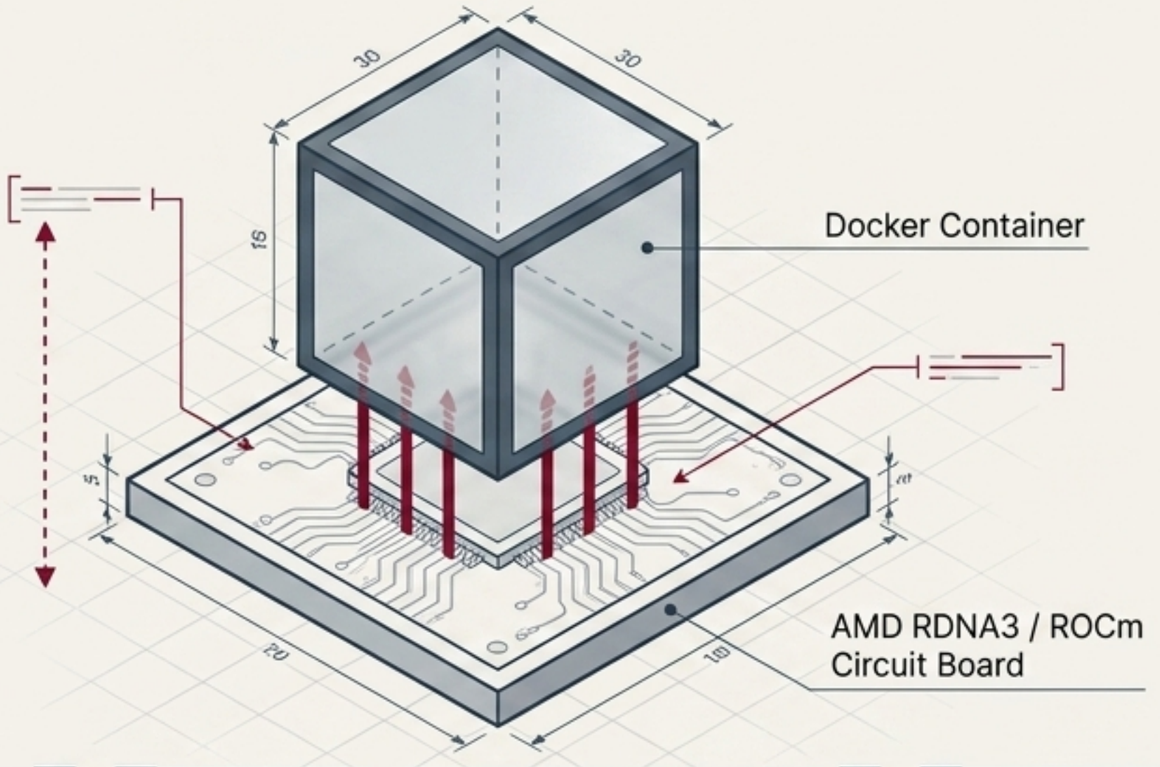
Secure interface utilizing SHA-256 hashed authentication. Empowers admins to view global usage statistics, issue refunds, and track per-model request counts.



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One-command deployment with deep AMD hardware passthrough



Base Image Foundation

Utilizes rocm/pytorch:rocm6.2_ubuntu22.04 ensuring native support for RDNA3 architectures (RX 7000 / RX 9070) without requiring compatibility hacks.

Frictionless Boot

Brought online with a single docker `compose up --build`.

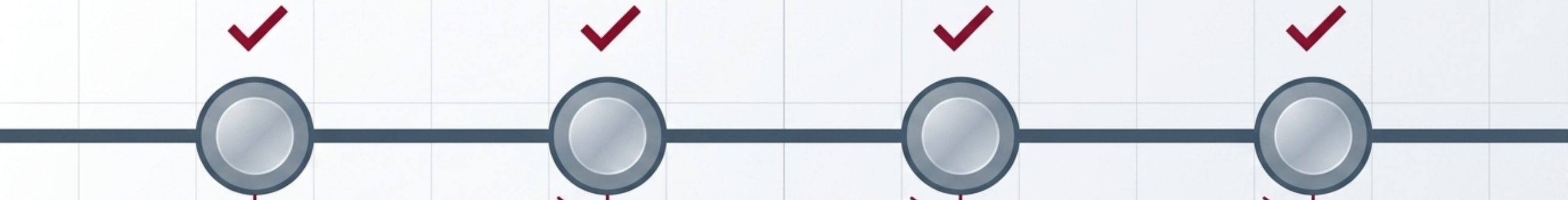
Direct Passthrough

Docker Compose configuration explicitly binds `/dev/kfd` and `/dev/dri`, granting the container bare-metal access to AMD GPU compute capabilities.

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Automated CI/CD gates ensure secure, verified builds on every commit



Gate 1: Hadolint Scanning

Enforces Dockerfile best practices, intentionally failing the build on warnings (e.g., unpinned `apt-get` versions). e.g., `MKO-POWER-9500`

Gate 2: Trivy CVE Analysis

Scans the final built image, aborting the pipeline if any `CRITICAL` severity vulnerabilities with **published fixes** are detected.

Gate 3: Layered Build

Utilizes GitHub Actions layer caching to rapidly compile the Python and OS dependencies.

Gate 4: E2E Smoke Tests

Automatically boots the stack via Compose, verifying `HTTP 200` routing, `valid JSON` database returns, and container health checks before passing.

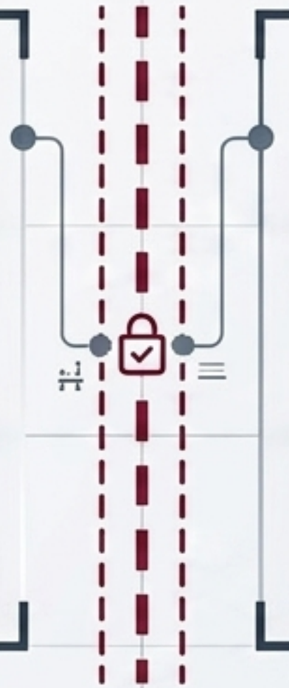
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The ‘No-Secrets’ architecture guarantees safe code distribution

Committed to Repository (Public)

- All `.py`, `.js`, `.css` application logic.
- `Dockerfiles` and `.github` workflows.
- `.env.example` with safely emptied values.



Firewalled / Gitignored (Local Runtime)

- Actual `.env` configuration files.
- The entire `data/` volume (User states, local API keys via UI, Qdrant vectors).

The Result

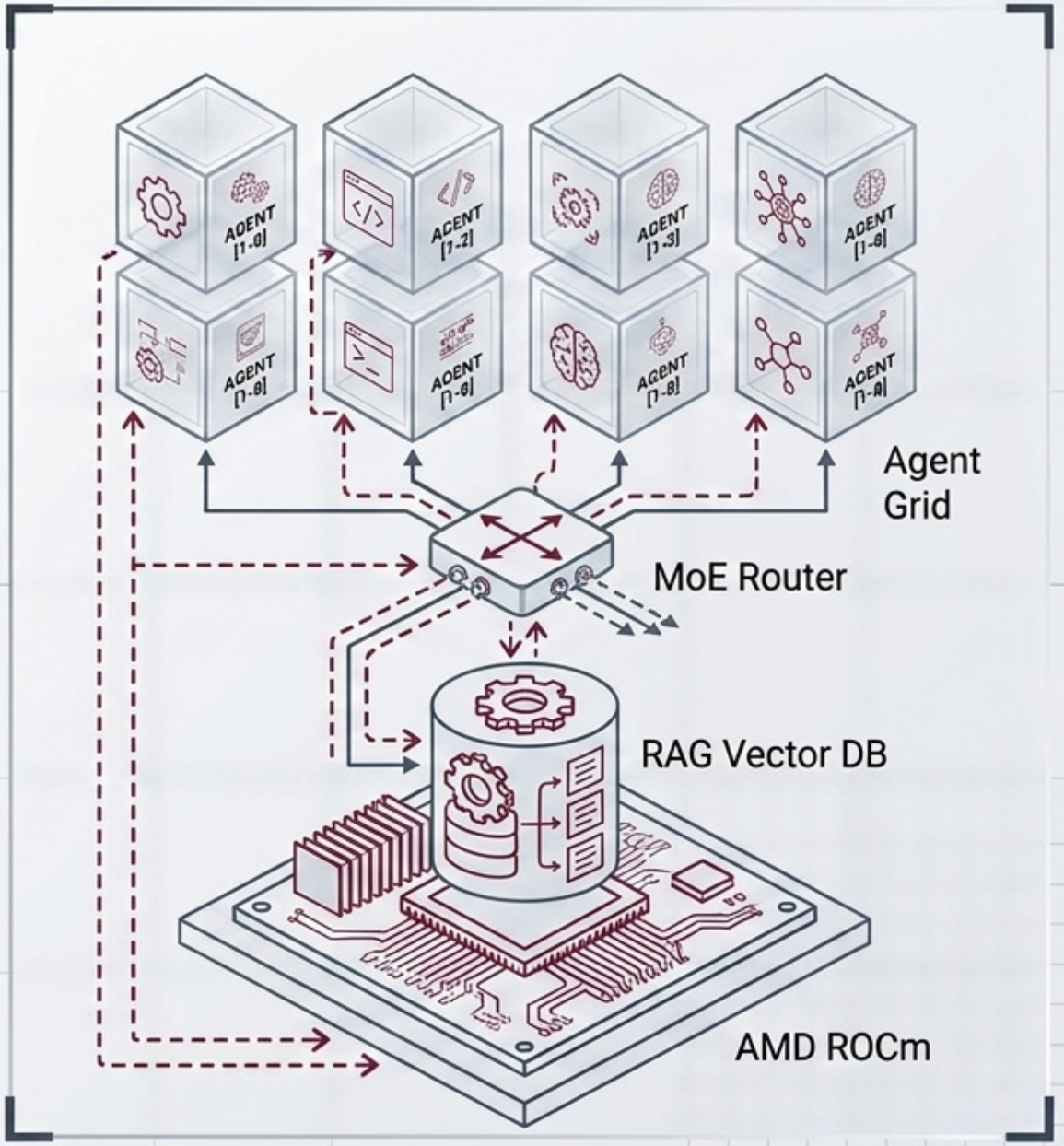
The full UI, admin panel, billing framework, and local agents (via Ollama) function flawlessly right out of the box without requiring a single API key.

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The most complete local orchestration framework available

MKO flawlessly unites local AMD hardware power with advanced multi-agent routing. It is a secure, secret-free, extensible architecture ready to deploy in the enterprise today.

```
docker compose up --build
```



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